姓名：_____ 学号：_____

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学号：

学院：

上海科技大学

2021-2022 学年第二学期本科生期末考试卷

开课单位：上海科技大学生命学院

授课教师：

考试科目：细胞生物学

课程代码：BIO1005

考生须知：

1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 参加闭卷考试的考生，除携带必要考试用具外，书籍、笔记、掌上电脑和其他电子设备等物品一律按要求放在指定位置。
3. 参加开卷考试的考生，可以携带教师指定的材料独立完成考试，但不准相互讨论，不准交换材料。

考试成绩录入表：

题目	1	2	3	4	5	6	7	8	9	10	总分
计分											
复核											

评卷人签名： **复核人签名：**

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1. 要求评卷人和复核人不能是同一人。
2. 试卷内页和答题纸编排格式由各学院和出题教师根据实际需要自定，每页须按顺序标注页码（除封面外），要求排版清晰、美观，便于在页面左侧装订。为方便印刷归档，建议使用 A4 双面印刷（学校有印刷一体机提供）。
3. 主考教师编写试卷时尽可能保证试题科学、准确、合理，如考试过程中发现试题有误，主考教师需负责现场解释，此类情况学校将作为教学评估记录的一部分。

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Part 1. Single-choice.

One answer per question (1 point per question, total 20 points)

1. In which organelle does respiration occur?
   A. Endoplasmic Reticulum
   B. Golgi Apparatus
   C. Chromosome
   D. Mitochondria
   E. Chloroplast

2. Which of the following description on embryonic stem cells is correct?
   A. They are made in the bone marrow.
   B. They are generated by meiosis.
   C. They are capable of developing into many, but not all, of the body's cell types.
   D. They are formed during gastrulation.

3. What is the definition of cell differentiation?
   A. Process by which a less specialized cell becomes a more specialized cell part
   B. Cells that are able to differentiate (change) into different cell parts
   C. When the whole cell is fully developed
   D. Process by which stem cells divide to make more stem cells

4. Which of the following is NOT generated during respiration?
   1. ATP
   2. H_2O
   3. CO_2
   4. O_2

5. Embryonic Stem (ES) cells are (_____)
   A. Pluripotent
   B. Totipotent
   C. Monopotent
   D. Multipotent

6. (_____)
During hematopoiesis which of the following cell types are generated?
   A. Goblet cells
   B. Lymphocytes
   C. Fibroblasts
   D. Cartilage

7. (_____) allow the passage of small molecules between adjacent cells.
   A. Desmosomes
   B. Tight junctions
   C. Gap junctions
   D. Hemidesmosomes

8. The nuclear lamina contains which of the following:
   A. Intermediate filaments
   B. Clathrin
   C. Microfilaments
   D. Tubulins

9. (_____) is the connection between cells that holds them together.
   A. Desmosomes
   B. Hemidesmosomes
   C. Adhesion junction
   D. Tight junctions
   E. Gap junctions

10. Cells adhere to Extracellular Matrices using (_____).
   A. Desmosomes
   B. Hemidesmosomes
   C. Adhesion junction
   D. Tight junctions
   E. Gap junctions

11. Which of the following molecules requires Ca^{2+} to maintain intermolecular adhesion?
   A. E-cadherin
   B. Tubulin
   C. Mucin
   D. Shh
   E. Connexins

12. Gap junctions are formed by (_____), and each hexameric (contains 6 monomers) complex is called a (_____)?

A. the fusion of two plasma membranes; connexons
B. protein hooks that extend into the membrane of adjacent cells; connexins
C. the insertion of transmembrane protein complexes that connect the cytoplasms of cells; connexons
D. gaps in the cell walls of plants; connexins
E. none of the above

13. The post-translation modification that marks proteins for proteolysis by the proteosome.

A. Phosphorylation
B. Methylation
C. Ubiquitination
D. Sumolyation
E. None of the above

14. Intracelluar proteins that facilitate vesicle fusion with organelle membranes or the plasma membrane, they are found on vesicle membranes and on target membranes.

A. Proteosomes
B. SNARE proteins
C. Rab proteins
D. Dynamin
E. Clathrin

15. The **DNA** macromolecule does not contain with of the following?

A. Nucleotides that contain ribose sugars
B. Nucleic Acids
C. Amino acids
D. Phosphate
E. Adenosine

16. The SERCA2 pump brings Ca^{2+} into the Endoplasmic Reticulum by (_____)

A. Active transport
B. Passive transport
C. Exocytosis
D. Diffusion along a gradient

17. Voltage difference across a membrane due to the slight excess of positive ions on one side and negative ion on the other is termed (_____)
    A. Action potential
    B. Membrane potential
    C. Depolarization
    D. Hyperpolarization
    E. None of the above

18. The process of a cell releasing vesicle cargo into the extracellular space is (_____)
    A. Phagocytosis
    B. Exocytosis
    C. Pinocytosis
    D. Diffusion
    E. Autophagy

19. Transport of water across the plasma membrane involves (_____).
    A. Passive transport
    B. Active transport
    C. Ligand-gated channels
    D. Mimicking ion caging

20. (_____) is protein with one or more oligosaccharide chains covalently linked to amino acid side chains.
    A. Protein glycosylation
    B. Glycoprotein
    C. GPI anchor
    D. ER signal sequence

Part 2. Multiple choice.

More than one answer per question (1 point each, total 10 points).

1. In which organelle(s) are proteins translated?
    A. Mitochondria
    B. Nucleus
    C. Golgi
    D. Endoplasmic Reticulum

2. Which of the following are part of the definition of the Cell Theory?

A. All forms of life contain one or more cells.
B. All functions that make life possible occur within cells.
C. Viruses are the smallest unit of life.
D. All cells come from pre-existing cells.
E. Cells possess the genetic material required to regulate all cell functions.

3. Enzymes perform which of these functions: (_____)
A. They increase the activation energy between reactants and products.
B. They increase up the rates of a chemical reactions in cells
C. They overcome larger activation energy thresholds.
D. That they enable energetically unfavorable reactions.
E. That they lower the activation energy between reactants and generation of products.

4. The driving forces for ion movement across a phospholipid membrane are due to:
A. Chemical gradient
B. Catalysis
C. Electrical (charge) gradient
D. Active transport
E. Osmosis

5. The purpose(s) of the electron transport chain in mitochondria are
A. To convert photon energy into electron transport.
B. To generate a hydrogen gradient across the inner membrane.
C. To convert CO_2 into a carbohydrate macromolecule.
D. To generate O_2
E. To convert chemical energy into the proton-motive force

6. The ribosome (_____)
A. Consists of 2 subunits
B. Is responsible for translating mRNAs
C. Is responsible for generating macromolecules of amino acids
D. Consists of RNA and proteins
E. Generates tRNAs

7. What is/are the following(s) NOT the function of the Golgi apparatus?
A. Transportation of protein molecules
B. Collection of protein molecules
C. Distribution of protein molecules
D. Digestion of protein molecules
E. Degradation of protein molecules

8. Which of the following factor(s) are the second messengers that are used for signal transduction? (_____)
   A. cAMP
   B. cGMP
   C. Sodium ion
   D. Calcium ion
   E. IP_3
   F. DAG
   G. Small G-proteins

9. Which of the following are true for Arp2/3?
   A. Involved in the formation of tight parallel bundles.
   B. Establishing new branches of microfilaments
   C. Binds to established microfilaments
   D. Involved in the formation of dendritic network of microfilaments
   E. Mimic actin monomers

10. Which are functions of the Endoplasmic Reticulum?
   A. Give the cell mechanical strength.
   B. Anchor/Support the organelles.
   C. Glycosylate proteins.
   D. Translate proteins that will be embedded into phospholipid membranes.
   E. Translate proteins that will undergo exocytosis.

Part 3. Matching (1 point each, total 10 points).

(A) Proteosome	A product of photosynthesis.
(B) Lateral inhibition	Degrades proteins.
(C) Compaction	The mechanism that determines whether an epithelial (ectodermal) cell will become a neural cell be prevented from becoming a neuroblast (neural cell).
(D) Cell death	The phenomenon that generates tightly adherent cells in the early embryo to form the first signs of cell polarity.
(E) Wnt	Generates two second messengers: one that remains embedded in the plasma membrane and the other can diffusion in the cytosol.
(F) Synaptonemal complex	A signal transduction pathway that uses proteolysis to promote transcription.
(G) Phospholipase C	Involved in generating outer surface of the skin.

(H) SHH(Sonic hedgehog)	Facilitates crossover between two non-sister chromatids Genes that require both copies to be mutated for a cell to acquire an overgrowth state
(I) Immunosurveillance	A signal transduction pathway that prevents proteolysis to promote transcription.
(J) Carbohydrate	The mechanism in which the immune system actively eliminates cancerous cells.

Part 4. Fill in the Blank (11 points).

Complete the sentence using the correct word or phrase.

1). The three enzymes responsible for ubiquitinating proteins are (_____, _____, and _____)? (3 points)

2). ATP synthase is which type of pump? (_____)
(1 point)

3). The electron transport chains in chloroplasts and mitochondria generate Hydrogen gradients across a lipid bilayer membrane to generate (_____). (1 points)

4). Initiator caspases and (_____) are the two groups of caspases that are involved in programmed cell death. (1 points)

5). The addition of actin monomers on one end of actin filament at the same rate as the loss results in (_____). (1 point)

6). Generation of microfilaments requires the (_____) nucleotide to polymerize. (1 point)

7). The cytoskeleton in the nucleus consists of (_____). (1 point)

8). (_____) can be phosphorylated and/or methylated on their N-terminal sequences to regulate gene transcription. (1 point)

9) Plants use photosynthesis to convert light energy to oxygen and carbohydrates. The first 3 stages of photosynthesis are called (_____).

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Part 5. True or False (1 point per question 10 points).

1. Ubquitination is a type of post-translational modification that always results in protein degradation.

2. Endocytosis is a process involving fusion of secretary vesicles or recycling vesicles with the plasma membrane.

3. Cancer is caused by the loss of function of both oncogenes and tumor suppressor genes.

4. Kinetochore assembles on the centromere and links the chromosome to microtubules from the mitotic spindle during cell division.

5. Cancer requires both copies of tumor suppressor to be mutated for a cell to acquire an overgrowth state.

6. Actin is a component of cytoskeletal elements.

7. Plectin connects microfilaments with Intermediate filaments.

8. In cytokinesis, local activation of RhoA triggers tubulin to form contractile rings to separate the mother cell into two daughter cells.

9. Cohesin subunits promote chromosome crossover in early mitosis.

10. Gastrulation is the phenomenon that generates tightly adherent cells in the early embryo to generate the first signs of cell polarity.

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Part 6. Short Questions (Choose 3 out of 6, 12 points)

Question 6A: Diagram and correctly label the organization of the nucleolus and how it is region is involved in the generation of a ribosome.

Question 6B: In muscle cells, a rise in intracellular Ca^{2+} triggers the contraction of the myosin–actin system. Meanwhile, muscle cells have specialized Ca^{2+} channels—called ryanodine receptors due to their sensitivity to the drug ryanodine—which are located in the membrane of the sarcoplasmic reticulum. In contrast to the IP_3-gated Ca^{2+} channels, Ca^{2+} itself can also open ryanodine receptors. Describe the roles and consequences of ryanodine channels in muscle cell contraction.

Question 6C: Here are two protein kinases, Kinase 1 and Kinase 2, which function sequentially in an intracellular signaling pathway. If either kinase contains a mutation that permanently inactivates its function, no signal transduction is seen in cells upon receiving an extracellular signal. A different mutation in Kinase 1 makes it permanently active, so that cells containing that mutation have a response even in the absence of an extracellular signal. In a double-mutant cell which contains Kinase 2 with the inactivating mutation and Kinase 1 with the activating mutation, you observe that the response is seen even in the absence of an extracellular signal. In the normal signaling pathway, does Kinase 1 activate Kinase 2 or does Kinase 2 activate Kinase 1? Explain your answer.

Question 6D: Platelet-derived growth factor (PDGF) can cause cancer when expressed inappropriately in an uncontrolled manner. Why do cancers not arise at wounds, where PDGF is released from platelets? Tell the differences and explain potential mechanisms.

Question 6E: The drug Taxol is just as pernicious to dividing cells as colchicine, and both are used as anti-cancer drugs. Firstly, Taxol binds tightly to microtubules and stabilizes them. When added to cells, Taxol promotes much of the free tubulin to assemble into microtubules. However, the drug colchicine prevents microtubule formation. Based on your knowledge of microtubule dynamics, please explain why both drugs are toxic to dividing cells despite their opposite actions.

Question 6F: When the skin gets injured, leukocytes are recruited to the injury site. Among those cells, neutrophils arrive first within a few minutes followed by monocytes and lymphocytes. How do these free-flowing leukocytes leave blood stream?

Part 7. Essay Questions (Choose 3 out of 5)

(Total 27 points)

Question 7A) (9 points).

Diagram and correctly label the complete process of protein translation from an mRNA transcript.

Question 7B (9 points).

Diagram and correctly label the complete action potential of a neuron. Include the different open and closed states of the associated voltage-gated Na+ channel.

Question 7C (9 points).

Accurately diagram and correctly label the complete process that results in the generation of cAMP by a ligand for G-protein coupled receptors.

Question 7D (9 points).

Accurately diagram and correctly label the processing and activation of Notch signaling pathway.

Question 7E) (9 points).

Fibronectin is a large glycoprotein found in all vertebrates and important for many cell-matrix interactions. When cells are seeded on fibronectin coated surface, they will adhere to the surface and spread. This attachment is required for cell growth, proliferation and survival. What signaling mediates cell adhesion to fibronectin? Describe the characteristics and all the steps of this signaling activation.

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Extra Sheet of paper for more answer space: Make sure you clearly <u>indicate which question you are answering</u> on this page!!!

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